VENKATESH SHANMUGAM

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LinkedIn | GitHub | Portfolio

SUMMARY

Senior AI/ML Engineer with 5+ years designing and deploying production ML systems. Solo-architected an enterprise ML platform with agentic LLM orchestration, knowledge graph reasoning (Neo4j), and real-time scoring pipelines on Databricks. Deep learning model development (78% ROC-AUC), distributed training, inference optimization (70% LLM cost reduction), and scalable data pipelines on 500M+ record datasets.

TECHNICAL SKILLS

ML / AI: PyTorch, TensorFlow, scikit-learn, Keras, PEFT/LoRA, Hugging Face, Transformers, DeepSpeed, LangChain, LangGraph
Languages: Python, SQL, C++, Java, R, Bash
Cloud & MLOps: AWS (SageMaker, S3, Glue, Athena, EC2), GCP (Vertex AI), Databricks, Delta Lake, MLflow, Docker, Kubernetes, CI/CD
Data & Infrastructure: Apache Spark, Kafka, Airflow, Neo4j, Elasticsearch, FastAPI, FHIR APIs, ETL Pipelines, A/B Testing
Specializations: Agentic AI, Knowledge Graphs, GraphRAG, Retrieval-Augmented Generation, NLP, Deep Learning, Computer Vision, Federated Learning, Distributed Training, NeMo Guardrails

PROFESSIONAL EXPERIENCE

AITHON AI | *Senior AI Engineer* Oct 2025 – Present
New York, NY

- Solo-architected the full ML/AI layer of an enterprise intelligence platform on **Databricks + Delta Lake**, building feature extraction pipelines across diverse structured and unstructured sources (SEC 10-K filings, CRM, market signals) with automated scoring, classification, and embedding generation
- Built a **graph-based ML reasoning engine (Neo4j + LangGraph)** with multi-hop inference and entity resolution, connecting latent signals across heterogeneous data sources to surface predictions that eliminated 20+ hrs/week of manual analysis per team
- **Reduced LLM inference costs by ~70%** by redesigning the model scoring architecture into a 3-pass pipeline with intelligent caching and selective re-ranking, cutting hundreds of redundant API calls per batch while improving prediction accuracy
- Designed **agentic ML pipelines (LangGraph)** with self-evolving cluster models that ingest, classify, and connect real-time signals using NLP extraction, topic modeling, and dynamic re-scoring for automated anomaly detection
- Led **LLM security and adversarial testing** on the production AI chatbot, identifying prompt injection vulnerabilities and schema disclosure risks, establishing security baselines for model deployment. Promoted to Senior AI Engineer within 4 months

ScriptChain Health | *Machine Learning Engineer* Jun 2025 – Sep 2025
Washington, DC

- Built a hybrid deep learning model (Graph Attention Network + time-aware LLM) for **30-day hospital readmission prediction, achieving 78% ROC-AUC** in clinical trials on an AI-powered metabolic syndrome platform
- Architected **agentic AI health platform using LangGraph** with multi-agent orchestration and NVIDIA NeMo Guardrails, automating personalized medical nutrition therapy aligned with AHA and U.S. Dietary Guidelines
- Optimized AWS infrastructure (S3, Glue, Athena) to reduce preprocessing time by **99% on a 500M+ record dataset**; cut GPU memory usage by 50%, saving ~$6K/year. Engineered HIPAA-compliant FHIR pipelines for real-time EHR data synchronization

The George Washington University | *Data Consultant* Aug 2024 – May 2025
Washington, DC

- Provided advanced statistical consulting (R, Python, SPSS, STATA) to researchers across public health, economics, and engineering. Led 5+ ML workshops training 100+ participants in model development and evaluation techniques

Tata Consultancy Services | *Digital Transformation Developer* Apr 2021 – Aug 2023
Chennai, India

- Designed and deployed RPA solutions across 14+ client teams, **reducing infrastructure costs by 40%** and improving process turnaround by 30%. Built 12+ automation bots integrated with CRM, billing, and network management systems

PROJECTS

Agentic Graph-RAG for Building Codes | Production-scale multi-agent system with 5 specialized LangGraph agents achieving 90% query accuracy improvement and 40% cost reduction with LangSmith observability and 99.9% uptime
AI-Text Discriminator | LoRA fine-tuned transformer on 1.2M+ text samples: 97% accuracy, 3x faster training, 50% reduced memory via parameter-efficient tuning (90% parameter reduction)

EDUCATION

The George Washington University, M.S. Computer Science, GPA: 3.88 / 4.0 Sep 2023 – May 2025
SRM University, B.Tech Computer Science, GPA: 3.5 / 4.0 Aug 2016 – May 2020